
NOTE

MVC and other design patterns are independent of any specific computer language and can be used in any language that supports OOP. If you're interested in learning more, I suggest that you search the web for OOP design patterns such as the Factory, Flyweight, Observer, and Visitor patterns; there are numerous video and text tutorials (as well as books) available on all of these. For a general introduction, Design Patterns: Elements of Reusable Object-Oriented Software (Addison-Wesley) by Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides (the Gang of Four) is considered the bible of design patterns.

Wrap-Up

When thinking about object-oriented programming, remember my initial definition of an object: data plus code that acts on that data over time.

OOP gives you a new way to think about programming, offering an easy and convenient way to group together data and code that acts on that data. You write classes and instantiate objects from those classes. Each object gets a set of all instance variables defined in the class, but the instance variables in different objects can contain different data and remain independent of each other. The methods of the objects can work differently because they're working on different data. Objects can be instantiated at any time and can be destroyed at any time.

When instantiating multiple objects from one class, you typically build a list or dictionary of objects, then later iterate over that list or dictionary, calling methods of each object.

As a final reminder, the three main tenets of OOP are:

Encapsulation Everything in one place, objects own their data.

Polymorphism Different objects can implement the same methods.

Inheritance A class can extend or modify the behavior of another class.